OverFlow: Flood Simulation and Satellite-Derived Flood Information for Scenario-Based Evacuation Planning in Laoag City

CAPSTONE PROJECT

JOHN HAROLD A. ALEJO
CYRUS KIRBY GAOR
BRADLEY KJIEL PASALO
JOSHEARIE KRISTOFFER LANDICHO



Department of Information Technology
COLLEGE OF COMPUTING AND INFORMATION SCIENCES
Mariano Marcos State University
City of Batac 2906 Ilocos Norte

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By

JOHN HAROLD A. ALEJO
CYRUS KIRBY GAOR
BRADLEY KJIEL PASALO
JOSHEARIE KRISTOFFER LANDICHO

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APPROVAL SHEET

This capstone project titled, **"OverFlow: Flood Simulation and Satellite-Derived Flood Information for Scenario-Based Evacuation Planning in Laoag City"**, prepared and submitted by JOHN HAROLD A. ALEJO, CYRUS KIRBY GAOR, BRADLEY KJIEL PASALO, JOSHEARIE KRISTOFFER LANDICHO in partial fulfilment of the requirements for the degree Bachelor of Science in Information Technology, has been examined and recommended for acceptance and approval for the oral examination.

Dr. Saturnina F. Nisperos
Adviser

PANEL OF EXAMINERS

Approved by the Committee on Oral Examination in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology on [EXAMINATION DATE TO BE CONFIRMED].

PANEL A. ONE
Chairman

PANEL B. TWO
Member

PANEL C. THREE
Member

Accepted in partial fulfillment of the requirements for the degree of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY.

_____
Date

Dr. Saturnina F. Nisperos
Dean

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The Researchers

ABSTRACT

Table of Contents

Title Page **i**

Approval Sheet **ii**

Acknowledgment **iii**

Abstract **iv**

Table of Contents **v**

List of Figures **vii**

List of Tables **viii**

List of Algorithms **ix**

1 Introduction **1**
- 1.1 Background of the Study 1
- 1.2 Statement of the Problem 2
- 1.3 Purpose and Objectives of the Study 2
- 1.4 Significance of the Study 3
- 1.5 Scope and Delimitations of the Study 3
- 1.6 Operational Definition of Terms 4

2 Review of Related Literature **5**
- 2.1 Introduction 5
- 2.2 Flood-Aware Road Accessibility and Geospatial Decision Support 5
- 2.3 Shelter Capacity and Population Allocation 6
- 2.4 Evacuation Simulation and Related Systems 7
- 2.5 Terrain Processing and Evacuation Behavior 8
- 2.6 Synthesis and Research Gap 9
- 2.7 Review of Related Systems 9
- 2.8 Review of Related Systems Matrix 13
- 2.9 Conceptual Framework 14

3 Methodology **16**
- 3.1 Research Design 16
- 3.2 Software Development Process 16

3.3 Locale and Participants 17
3.4 Data Sources and Gathering Tools 18
3.5 Research Instruments 19
3.6 Data Gathering and Analysis Procedures 20
3.7 Treatment of Data 21

4 Results and Discussion 22
4.1 System Design 22
4.2 Initial Prototype 32
4.3 Preliminary Flood-Data and Model Checks 32
4.4 System Evaluation 33

5 Conclusions and Recommendations 34
5.1 Conclusion 34
5.2 Recommendation 34

References 35

Appendices 39

Appendix A Detailed Tables 40

Appendix B Questionnaires 41

List of Figures

2.1 Las Nieves evacuation-center system dashboard (screenshot provided for this study; system described in [8]) 10
2.2 Iowa Routing Decision Support System interface (screenshot provided for this study; system described in [5]) 11
2.3 Evacuation Management System map interface (screenshot provided for this study; system described in [25]) 12
2.4 LifeSim project interface (screenshot provided for this study; system described in [24]) 13
2.5 Conceptual framework of the proposed OverFlow system 14
3.1 Iterative and incremental software development process for OverFlow 17
3.2 Geomorphological map of Laoag City, reproduced from Beroya and Aydin [40] 18
4.1 Proposed interface wireflow 23
4.2 Proposed system context 24
4.3 Proposed scenario sequence 25
4.4 Proposed high-level architecture 26
4.5 Proposed entity-relationship model 27
4.6 Proposed deployment 28
4.7 Proposed data flow 29
4.8 Proposed backend components 29
4.9 Proposed planning activity 30
4.10 Illustrative dashboard mockup; all displayed metrics are placeholders 31

List of Tables

2.1 Functions documented in related systems and proposed for OverFlow 14

List of Algorithms

1 Introduction

1.1 Background of the Study

Flood evacuation requires decisions that change with each event. The number of people needing assistance, the roads they can use, and the space available in evacuation centers may differ from one situation to another. In Laoag City, these decisions can involve communities across its 80 barangays [1]. The Department of Social Welfare and Development recorded 78 Laoag barangays and 45,498 people affected by the combined effects of the Southwest Monsoon and Super Typhoon Egay in 2023 [2]. These figures describe overall storm impacts, rather than the number of people flooded or evacuated. More recently, the Philippine Space Agency published a radar-derived map of possible flood extents from the August 2026 Southwest Monsoon [3].

Geographic records can help locate communities, roads, and flood-prone areas, but they do not directly answer planning questions. If a road becomes unusable, a center reaches capacity, or more residents require evacuation, planners need to know which destinations remain reachable and how many people cannot be accommodated. The Laoag datasets available for this study contain barangay boundaries, roads, population records, elevation, and flood-susceptibility information. These sources provide a basis for mapping, while evacuation demand, road access, and usable center capacity must be specified for a particular planning scenario.

These changing conditions need to be examined together. OverFlow is proposed to combine flood information with route and evacuation-center analysis. It will help planners compare alternatives, including cases where roads or listed centers cannot serve everyone.

1.2 Statement of the Problem

This study addresses the difficulty of determining feasible evacuation routes and center assignments when demand, road access, and center capacity change. Specifically, it asks:

1. How can available Laoag geographic, population, evacuation-area, and flood information be organized for planning across all 80 barangays?

2. How do changes in expected evacuees, road closures, and usable center capacity affect reachable routes, center assignments, and unserved demand?

3. How well does the proposed system perform its specified planning tasks, and how do intended users assess those tasks if staff participation is obtained?

1.3 Purpose and Objectives of the Study

The purpose of the study is to develop and evaluate a geospatial tool for preparing and comparing flood-evacuation scenarios in Laoag City. A planner will select a barangay, set the expected number of evacuees, identify closed roads, and enter the usable capacities of listed evacuation areas. The proposed system will then calculate routes by mapped road distance, assign people to reachable centers within capacity, and report those who remain unserved.

The study specifically aims to:

1. Assess and prepare the available Laoag data for scenario planning across its 80 barangays, including the identification of missing or uncertain inputs.

2. Design and develop an interface for viewing geographic and flood information, configuring evacuation conditions, and comparing scenarios.

3. Implement and test distance-based routing and capacity-limited center assignment under planner-specified road closures.

4. Examine the feasible uses and limitations of elevation data, barangay-based SFINCS flood simulation, and dated satellite radar flood maps within the proposed system.

5. Evaluate the planning functions through documented tests and, if participation is secured, tasks completed by intended users.

1.4 Significance of the Study

For disaster planners, the proposed system offers a means of examining how a change in road access or center capacity affects an evacuation arrangement. Its report of unreachable locations and unserved demand can draw attention to constraints that require further planning.

Laoag residents are the intended indirect beneficiaries of more thoroughly examined evacuation options. The study may also help identify which local data require verification or updating before a geospatial planning tool can be relied upon in practice.

1.5 Scope and Delimitations of the Study

The study covers evacuation-scenario planning for Laoag City's 80 barangays. Its proposed functions include barangay selection, entry of evacuation demand and usable center capacities, manual road closures, route and center analysis, geographic display, and scenario comparison. Assignments will use listed evacuation areas and available road connections. Where these data do not support an assignment, the result will be reported as a gap.

The digital elevation model (DEM) provides terrain for barangay-based SFINCS simulations. The model domain may extend beyond the selected barangay to represent water entering or leaving it. Flood-susceptibility data and satellite radar (SAR) detections provide separate reference layers. Given the available terrain and event data, simulated depths will be treated as exploratory, and road closures will remain a planner input.

The study is delimited to preparedness planning and scenario comparison. It does not provide real-time forecasts, emergency dispatch, public navigation, or operational estimates of vehicle requirements and clearance time. Routes are calculated by distance under entered closures; they do not establish travel speed or safety.

1.6 Operational Definition of Terms

Evacuation demand refers to the number of people expected to evacuate from a barangay in a specified scenario.

Usable capacity refers to the number of evacuees a listed evacuation area can accommodate under the conditions entered for that scenario.

Reachable route refers to a path through the mapped road network after the planner's specified closures have been applied.

Unserved demand refers to evacuees who cannot be assigned to a reachable evacuation area within its usable capacity.

DEM refers to a grid of elevation values used to represent terrain.

SAR flood extent refers to an area classified as possible surface water from satellite radar data acquired at a stated time.

SFINCS refers to the flood-simulation method selected for barangay-based scenarios.

2 Review of Related Literature

2.1 Introduction

Flood-evacuation planning connects hazard assessment, road access, shelter assignment, and human movement. This review examines how these concerns have been represented in geospatial systems and identifies the methods most relevant to a city-level planning tool.

2.2 Flood-Aware Road Accessibility and Geospatial Decision Support

Li et al.'s review of urban flood evacuation research places hazard assessment before shelter selection, route development, and movement analysis [4]. Alabbad et al. showed in two Iowa cities that inundation can disconnect roads and bridges, lengthen journeys, and change the nearest reachable emergency facility [5]. Together, these studies establish why route analysis must use the access conditions of a flood scenario rather than the ordinary road network alone.

GIS studies in Thailand and Nepal combine flood-susceptibility mapping with route and destination analysis [6, 7]. Parajuli et al. also accounted for the effects of slope and flooding on walking speed and compared their susceptibility map with satellite observations. In the Philippines, Bolanio et al. assessed the suitability of existing evacuation centers in Las Nieves and presented the results through a web map [8]. Abuzo et al. compared road conditions under normal and flood scenarios in Cagayan de Oro [9]. These cases show the value of examining destinations and their connecting routes together, using local geographic evidence.

Dynamic-route studies address changes that a static map cannot represent. Ogawa et

al. weighted roads according to present and projected inundation in a Japanese evacuation simulation [10]; Jeon et al. recalculated pedestrian routes when a flooded path or destination became unavailable [11]. Their predictive and real-time functions require hazard data and validation beyond a static road-closure scenario. OverFlow uses modeled exposure to inform planner review and excludes a road from routing when the planner confirms its closure.

2.3 Shelter Capacity and Population Allocation

Shelter choice involves more than proximity. Qiu et al. combined flood-risk assessment, shelter selection, and evacuation simulation, reporting improved outcomes for their modeled shelter arrangement [12]. Sharbaf et al. treated population exposure, shelter risk, and route risk as parts of one shelter-network problem [13]. Wang et al. compared routes and shelters under different road-risk and pedestrian-demand priorities [14]. These studies show that a nearby facility may be unsuitable because of exposure, access, or limited space.

A Philippine location-routing model by de Veluz et al. jointly considered facilities, routes, capacities, travel time, and vehicles across typhoon scenarios [15]. Gama et al. examined how shelter openings, evacuation orders, and assignments change over time as flood and road conditions develop [16]. Both studies demonstrate that assignments depend on the conditions and priorities of a particular scenario.

Other work examines who can reach a shelter and whether enough space is available. Bera et al. found that existing shelters in an Indian case could not serve all households within the tested travel-time limits [17]. Rendana et al. showed that areas classified as low flood risk are not necessarily suitable shelter sites once road access and other conditions are considered [18]. Yin et al. examined the transfer needs of older residents in coastal cities, while Janatabadi and Ermagun identified communities with high flood risk and low

shelter access [19, 20]. Chen et al. explicitly recorded demand that existing shelters could not accommodate [21].

For OverFlow, the initial assignment rule is deliberately explicit: evacuees are assigned only to reachable centers with available capacity, and remaining demand is reported. The literature identifies additional concerns, including shelter risk and mobility needs, that require local data and defined rules before inclusion in the allocation method.

2.4 Evacuation Simulation and Related Systems

Aketa et al. provided an early example of linking flood and road-network analysis in a GIS evacuation system [22]. Mizumura et al. later compared routes that leave inundated areas quickly with ordinary shortest paths, showing that shorter distance and lower flood exposure can lead to different choices [23]. The U.S. Army Corps of Engineers' LifeSim represents a more detailed approach, combining warnings, population movement, traffic, flood exposure, and potential life loss [24]. These systems illustrate a progression from route calculation to dynamic evacuation modeling, with increasing demands for behavioral and hazard evidence.

González-Villa et al. combined GIS, network analysis, and pedestrian and vehicle simulation in a wildfire evacuation system [25]. Du et al. coupled flood, agent-based, and traffic models to compare shelter arrangements and routing strategies; their results showed that shelter distribution and system-level routing can substantially affect modeled clearance time [26]. These systems demonstrate the value of comparing complete arrangements, while their traffic and behavior models require data that a simpler scenario tool does not assume.

Studies of data-constrained and coastal settings reinforce the importance of stating

what a simulation represents. Butters and Dawson tested shelter locations, crossings, and population samples in Kibera and found that a very small sample produced unstable risk estimates [27]. Yang et al. examined warning time, infrastructure failure, and shelter capacity in Shanghai [28]. Bustami et al. showed how compound hazards and local shelter placement changed modeled outcomes in Connecticut [29]. Their numerical results belong to those modeled settings; the transferable lesson is to compare alternatives under documented inputs and examine local as well as citywide effects.

Evacuation also places demands on vehicles and post-arrival supplies. Santana-Robles et al. linked shelter assignment with aid delivery after a Mexican flood [30]. Woo and Kang combined bus evacuation, congestion, shelter selection, and relief routing [31]. Insani et al. modeled vehicle reuse, uncertain demand, and relief delivery before inundation reached affected areas [32]. These studies motivate recording resource assumptions alongside routes and center assignments, without treating a simplified vehicle estimate as a traffic or logistics simulation.

2.5 Terrain Processing and Evacuation Behavior

O'Callaghan and Mark's D8 method derives drainage directions and contributing area from gridded elevation data [33]. It can provide a terrain reference, but flood depth and timing require a flood model. OverFlow uses SFINCS for rainfall-driven scenarios, with its inputs, forcing, and outputs documented by Deltares [34, 35, 36].

Observed behavior also differs from a mathematically shortest path. Lim et al. found that route choice among flood-affected households in Quezon City was influenced by hazard, socioeconomic, and evacuation factors [37]. Borowska-Stefańska et al. showed that self-evacuation can redistribute traffic well beyond a flood zone [38]. OverFlow's routes are

planning options under stated access and travel assumptions, rather than predictions of individual movement or congestion.

2.6 Synthesis and Research Gap

The literature establishes strong methods for flood mapping, road analysis, shelter assessment, and evacuation simulation, but these functions are often studied separately or within data-intensive models. Philippine studies provide evidence on route behavior, center suitability, flooded roads, and location-routing decisions [37, 8, 9, 15]. The reviewed local applications do not combine these concerns in one city-level interface for comparing flood and operational scenarios.

OverFlow addresses this gap by bringing flood-scenario information, road closures, evacuation demand, and reachable-center capacity into one planning workflow. Its evaluation will examine the flood model, route and assignment constraints, system functions, and staff use separately.

2.7 Review of Related Systems

The following systems provide concrete comparisons for OverFlow's proposed combination of flood information and evacuation planning. They differ in purpose, data requirements, and the decisions they support.

Las Nieves evacuation-center web map. Bolanio et al. developed a web application for viewing evacuation-center status and spatial capacity in Las Nieves, Agusan del Norte. Their study also assessed existing centers against a flood scenario and mapped site suitability. The system demonstrates the value of keeping local center records and hazard information accessible to planners; the reported application does not describe assigning evacuees through

a road network under changing center capacities [8].

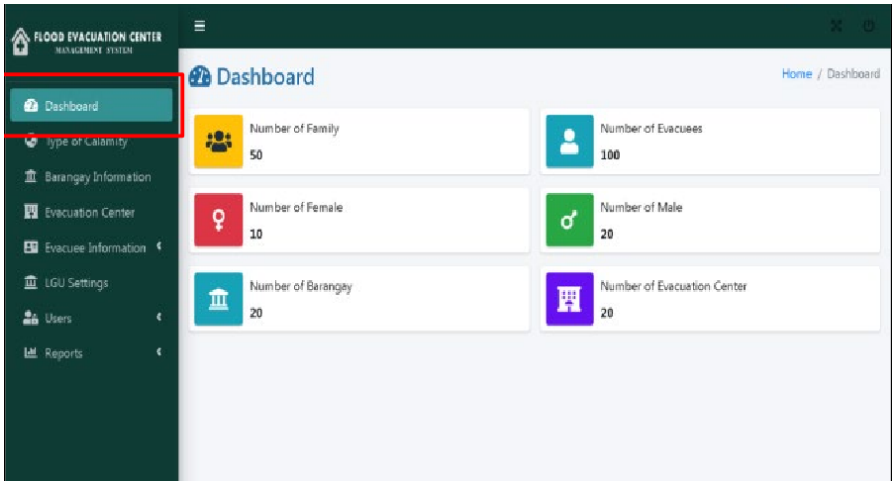


Figure 2.1: Las Nieves evacuation-center system dashboard (screenshot provided for this study; system described in [8])

Iowa Routing Decision Support System. Alabbad et al. developed a web application for comparing road conditions, routes to critical facilities, service areas, and possible emergency-facility locations under different flood extents. Its Iowa cases show how inundation can divide a road network and change reachable destinations. Its facility-location analysis selects sites for services, whereas OverFlow proposes assigning scenario demand to listed evacuation areas within entered usable capacities [5].



Figure 2.2: Iowa Routing Decision Support System interface (screenshot provided for this study; system described in [5])

Evacuation Management System. González-Villa et al. presented a GIS-based system that links assembly points, shelters, routes, and pedestrian and vehicle movement. An operator can revise conditions and run the evacuation again to compare strategies. Its reported case concerns a wildfire in Gran Canaria, so its movement estimates do not establish flood behavior or travel times for Laoag [25].

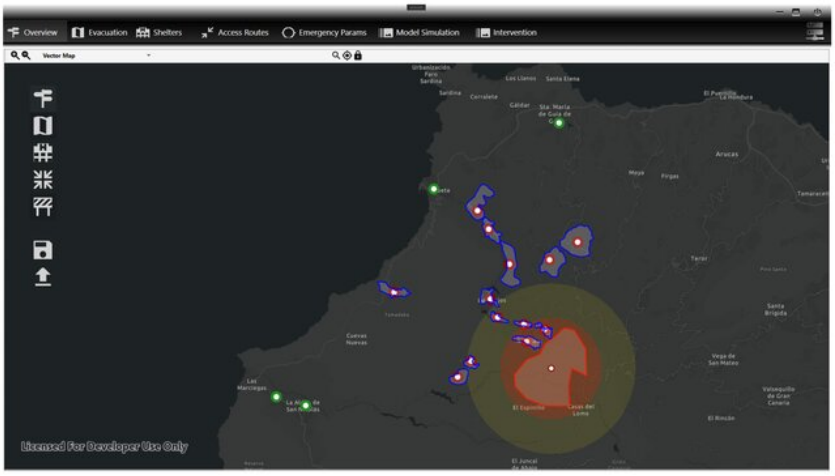


Figure 2.3: Evacuation Management System map interface (screenshot provided for this study; system described in [25])

LifeSim 2.0. The U.S. Army Corps of Engineers' LifeSim represents evacuation decisions and movement as flood conditions develop. It uses road, destination, population, warning, and hazard inputs to examine traffic and potential consequences. This provides a more detailed example of flood-evacuation simulation, but its behavioral and traffic requirements exceed the data available for OverFlow's initial distance-based analysis [24].

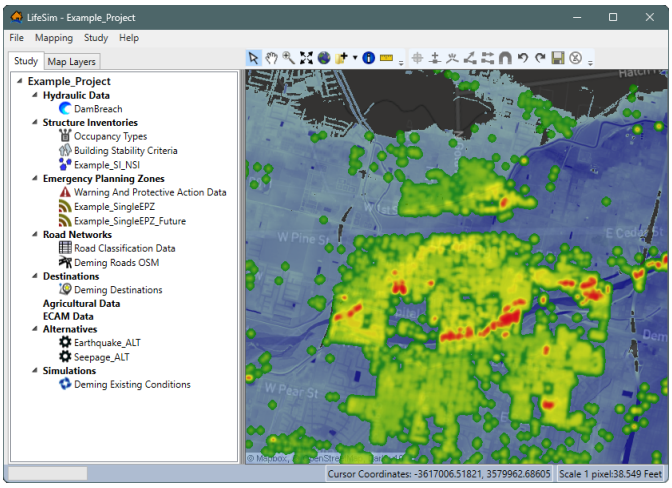


Figure 2.4: LifeSim project interface (screenshot provided for this study; system described in [24])

These systems show different ways to connect hazards, accessibility, destinations, and scenario analysis. OverFlow adopts a narrower planning task: for a selected Laoag barangay, it proposes a flood scenario and an explicit report of routes, center assignments, and demand that cannot be served under the planner's stated conditions.

2.8 Review of Related Systems Matrix

Table 2.1 compares documented functions of the reviewed systems with those proposed for OverFlow. The symbols are defined below the table.

The comparison highlights two choices in the proposed design. OverFlow will report capacity-limited assignments and unserved demand rather than treating the presence of a center as sufficient. It will also distinguish a SFINCS scenario from historical SAR detections and flood-susceptibility information so that each layer is interpreted according to its source and evidence.

Table 2.1: Functions documented in related systems and proposed for OverFlow

Function	Las Nieves web map	Iowa RDSS	EMS	LifeSim	OverFlow
Spatial hazard information	✓	✓	✓	✓	✓
Road routing or accessibility	X	✓	✓	✓	✓
Center or destination information	✓	✓	✓	✓	✓
Capacity-limited evacuee assignment	X	X	X	X	✓
Comparison of changed scenarios	X	✓	✓	✓	✓
Flood simulation within the planning workflow	X	X	X	X	✓

Note. ✓ = documented in a cited source, or proposed for OverFlow; X = not described in the cited source. Iowa RDSS = Iowa Routing Decision Support System; EMS = Evacuation Management System. Source descriptions: [8, 5, 25, 24].

2.9 Conceptual Framework

Figure 2.5 links the planning problem, the proposed analysis, and the results to be examined. The inputs combine prepared Laoag geographic records with values entered for a particular scenario. Terrain, rainfall, and boundary information support a SFINCS run focused on the selected barangay; the model area may extend beyond its boundary to represent water entering or leaving it. SAR and susceptibility layers provide separate hazard references. Road closures, expected evacuees, and usable center capacities govern the evacuation analysis.

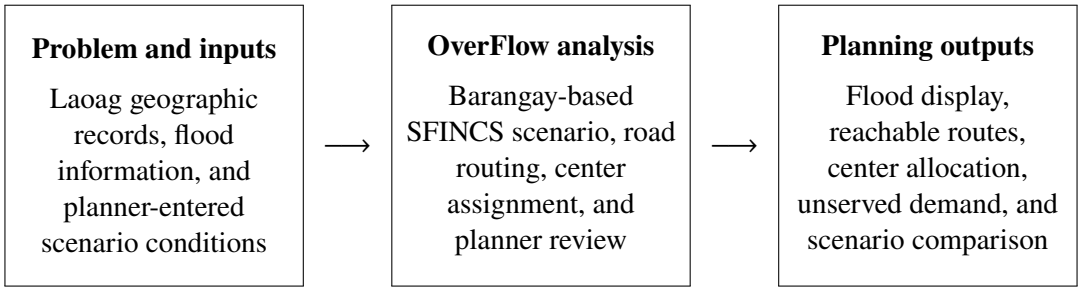


Figure 2.5: Conceptual framework of the proposed OverFlow system

The framework places the planner between the flood display and the road-access decision. A road is removed from route calculation only when the planner marks it closed. Chapter 3 describes how the data, model, and planning functions will be evaluated.

3 Methodology

This chapter describes how the study will prepare Laoag's geographic data, develop the proposed planning functions, and evaluate their results.

3.1 Research Design

The study will use a descriptive-developmental design organized through design-science research. The descriptive component will examine the available data and document local evacuation-planning practices if personnel participate. The developmental component will build and evaluate a geospatial tool that responds to those requirements. Design science structures the work from problem identification and solution objectives through development, demonstration, evaluation, and communication [39].

3.2 Software Development Process

Within the design-science study, OverFlow will be built through iterative and incremental software development. Each iteration will plan, design, implement, test, and review a set of functions. Feedback can revise the current increment before work proceeds to the next one. The planned increments add Laoag maps and barangay selection; scenario entry, routes, and center assignment; barangay-based flood simulation and satellite-derived reference layers; then scenario comparison. Figure 3.1 shows this cycle. The initial synthetic prototype demonstrates the interaction; later increments will use prepared Laoag data.

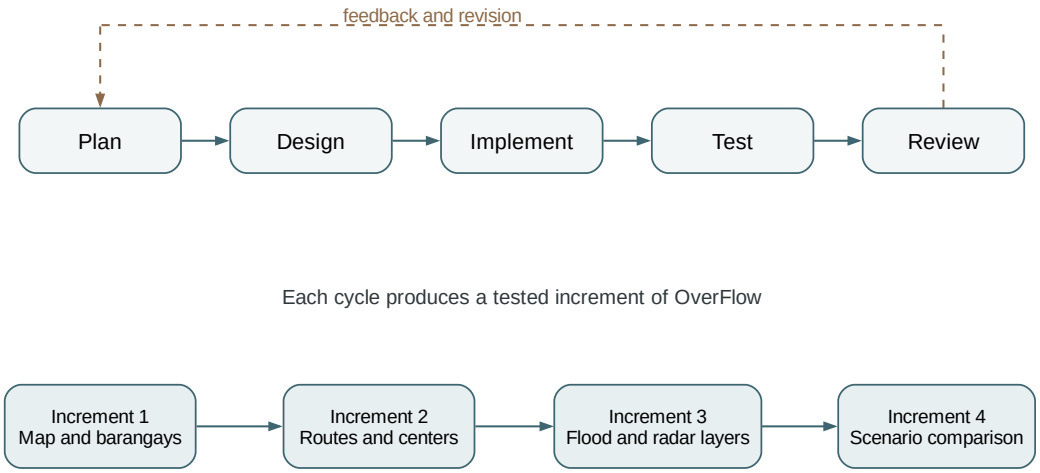


Figure 3.1: Iterative and incremental software development process for OverFlow

3.3 Locale and Participants

The study area is Laoag City, Ilocos Norte. The planning interface will include all 80 barangays. Each flood scenario will focus on a selected barangay, with the SFINCS domain extending beyond its boundary where required to represent incoming or outgoing water. Figure 3.2 shows the city's channels and landforms in an earlier geomorphological study; the barangay boundaries used for OverFlow will come from the Laoag City Government data.

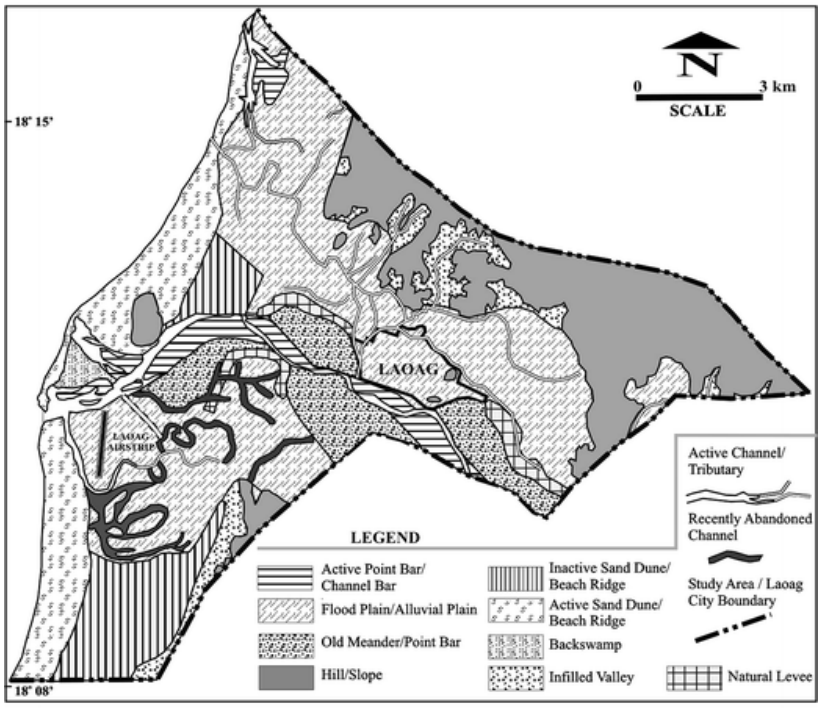


Figure 3.2: Geomorphological map of Laoag City, reproduced from Beroya and Aydin [40]

Personnel involved in evacuation planning or response will be invited purposively for interviews and task-based evaluation, subject to institutional permission and ethics requirements. The Laoag City Disaster Risk Reduction and Management Office is a proposed source of participants. The final number and roles of participants will be reported after recruitment.

3.4 Data Sources and Gathering Tools

The study will examine the barangay boundaries, roads, population records, elevation grid, and flood-susceptibility layer received from the Laoag City Government. Philippine Statistics Authority barangay population data will provide a dated population reference [1].

Listed evacuation areas will be checked for location and status; until usable capacities are verified, capacities entered by a planner will remain scenario assumptions.

The historical flood layer will come from the Philippine Space Agency's August 31, 2026 SAR product [3]. Its acquisition time, coverage, classes, and known detection limits will be recorded. A location without a SAR detection will not automatically be classified as dry.

GIS and programming tools will be used to inspect coordinate systems, align layers, check missing data, and prepare a connected road network. An inventory will retain each dataset's source, date, coverage, resolution where applicable, and restrictions on use.

3.5 Research Instruments

A data-inventory form will document whether each source is suitable for its proposed use. A semi-structured interview guide will ask participating personnel how they identify affected communities, determine road access, select evacuation areas, and review alternative plans. The guide will provide consistent topics while allowing participants to explain practices not anticipated by the researchers.

A system-test sheet will record each test's inputs, expected result, observed result, and outcome. For staff evaluation, a task sheet will direct participants to select a barangay, change demand, close a road, adjust a center's capacity, and compare scenarios. Task completion will show whether the workflow can be carried out. A brief questionnaire and open comments will separately capture participants' judgments about the clarity and usefulness of the results. The instruments will be reviewed for clarity and alignment with the study objectives before administration.

3.6 Data Gathering and Analysis Procedures

The researchers will first secure the permissions required to use the datasets and to conduct any activities involving personnel. Interviews and staff tasks will proceed only after the applicable ethics review and informed consent. Participation and findings will be reported without identifying individual staff members.

Geographic records will then be checked and prepared. Barangay boundaries and population will be matched by code. Road lines will be examined for missing segments, intersections, and connectivity. Each barangay used in an analysis will have a documented access point on the road network. Evacuation areas will be included only when their identity and location can be established; unverified capacity will be clearly identified as a scenario input.

For each scenario, the planner will enter expected evacuees, usable center capacities, and closed road segments. Closed segments will be removed from the traversable network. Routes to reachable centers will be calculated using mapped road length as the weight. Center assignment will first serve as many evacuees as the network and capacities permit, then favor shorter total road distance among feasible assignments. The output will identify unreachable origins, full centers, and unserved demand. These calculations will use aggregate evacuee counts and will not estimate vehicle schedules or travel time.

Elevation, flood susceptibility, and SAR will be displayed as distinct sources of information. SFINCS simulations will use prepared terrain, rainfall, and boundary inputs, followed by numerical checks [34, 35, 36]. Local flood accuracy will be assessed only if suitable dated observations and event inputs are available. Road closures will remain a planner input.

Functional tests will use controlled cases with known routes and capacities to check closures, reachability, assignment, and unserved demand. Selected cases will then be repeated with prepared Laoag data to identify data-related gaps. If staff participation is obtained, participants will complete the task sheet and provide ratings and comments after using the system.

3.7 Treatment of Data

Dataset findings will be reported by source and type of limitation. Functional tests and staff tasks will be summarized as counts and percentages, with the number attempted stated. Scenario results will report route distance, assigned evacuees, capacity use, and unserved demand under the stated inputs.

Questionnaire responses will be presented as frequencies and item-level means on the five-point scale, with the number of responses for each item. Interview responses and open comments will be grouped into themes relevant to planning requirements and system use.

4 Results and Discussion

This chapter presents the system design, the initial interaction prototype, and preliminary Laoag flood-data and model checks.

4.1 System Design

The proposed workflow begins when a planner selects one of Laoag City's 80 barangays and defines a scenario. The planner enters rainfall assumptions and expected evacuees, identifies usable evacuation areas and their capacities, and marks any roads that should be treated as closed. SFINCS is the selected method for simulating flooding in a bounded area around the chosen barangay. The model area may extend beyond the administrative boundary where needed to represent water entering or leaving it, while the scenario results focus on the selected barangay.

The interface wireflow in Figure 4.1 traces the planned movement from barangay selection and scenario entry to map viewing and result comparison. The sequence does not require the user to draw an analysis boundary; selection is based on prepared barangay polygons.

The context diagram in Figure 4.2 places the planner, geographic records, and potential external data sources around OverFlow.

Figures 4.3 and 4.4 show the proposed exchange between the interface, scenario processing, flood simulation, and evacuation analysis. They distinguish the data entered for a particular scenario from the geographic records prepared for repeated use.

The data model in Figure 4.5 organizes barangays, roads, evacuation areas, scenarios, and results. Its separation of source records from scenario values allows a planner to change

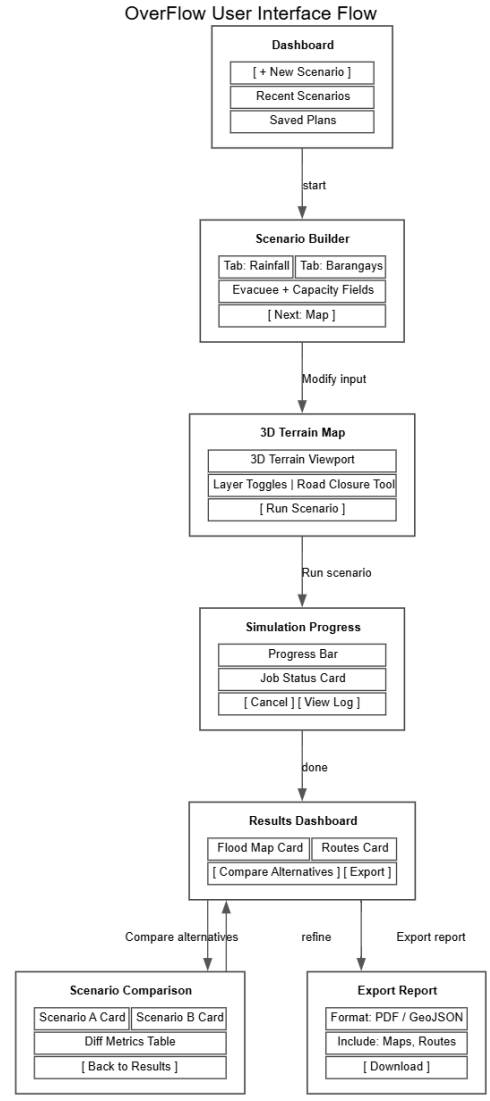


Figure 4.1: Proposed interface wireflow

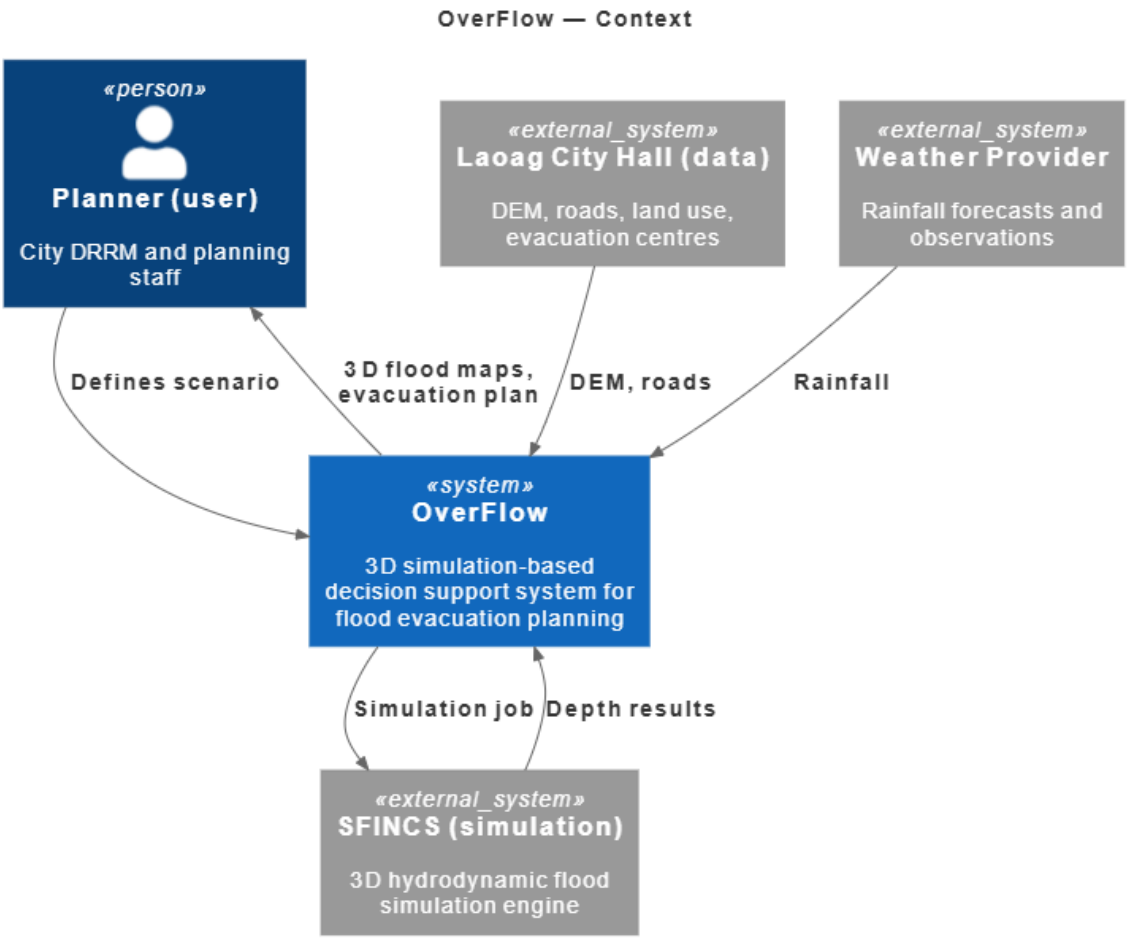


Figure 4.2: Proposed system context

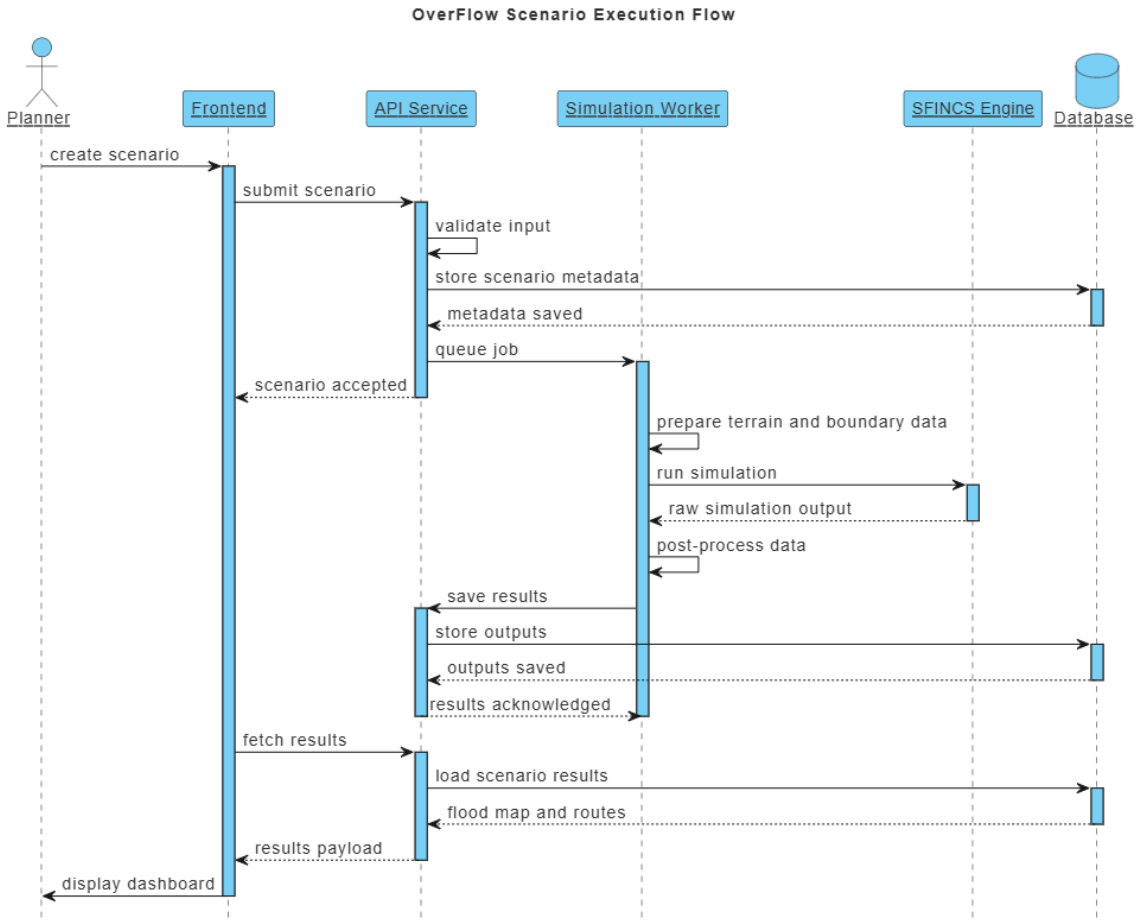


Figure 4.3: Proposed scenario sequence

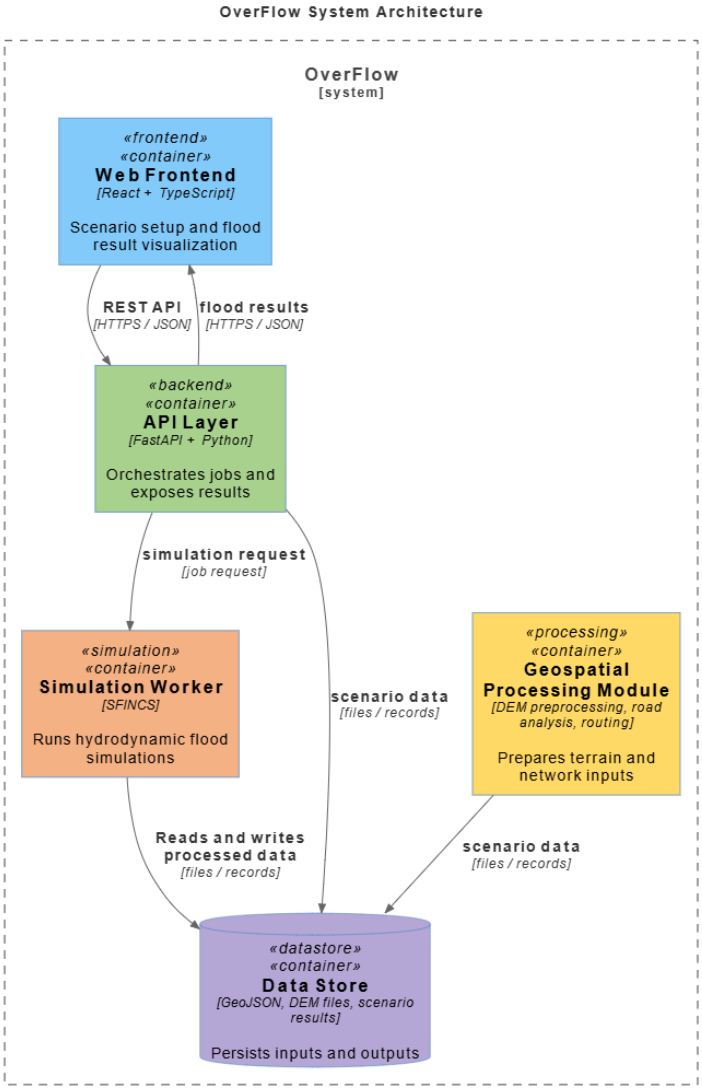


Figure 4.4: Proposed high-level architecture

demand or usable capacity without treating those assumptions as permanent facts about Laoag.

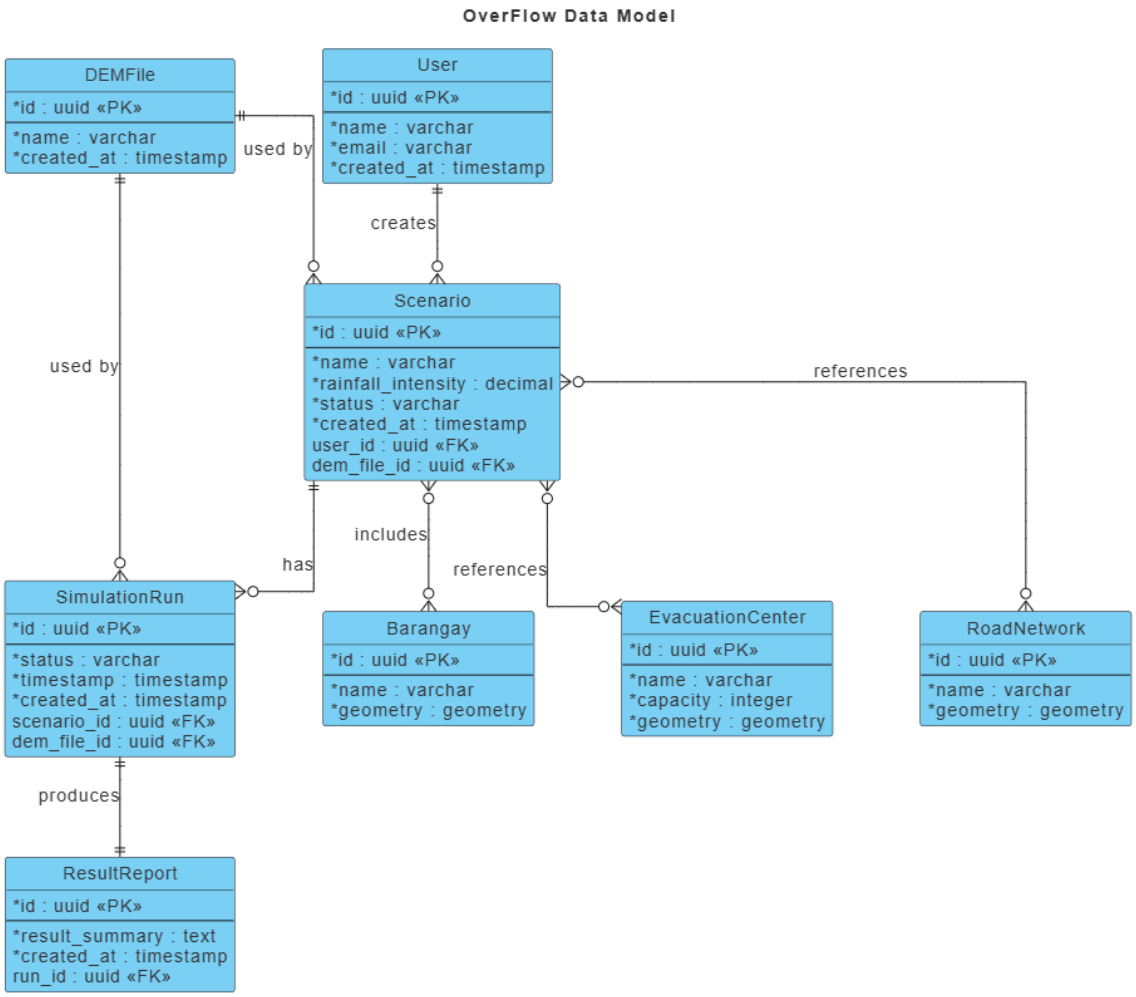


Figure 4.5: Proposed entity-relationship model

Figure 4.6 shows the intended arrangement of the interface, processing services, and data storage.

The data-flow and backend diagrams in Figures 4.7 and 4.8 trace how prepared geographic inputs and planner-defined conditions would produce flood displays, routes, center

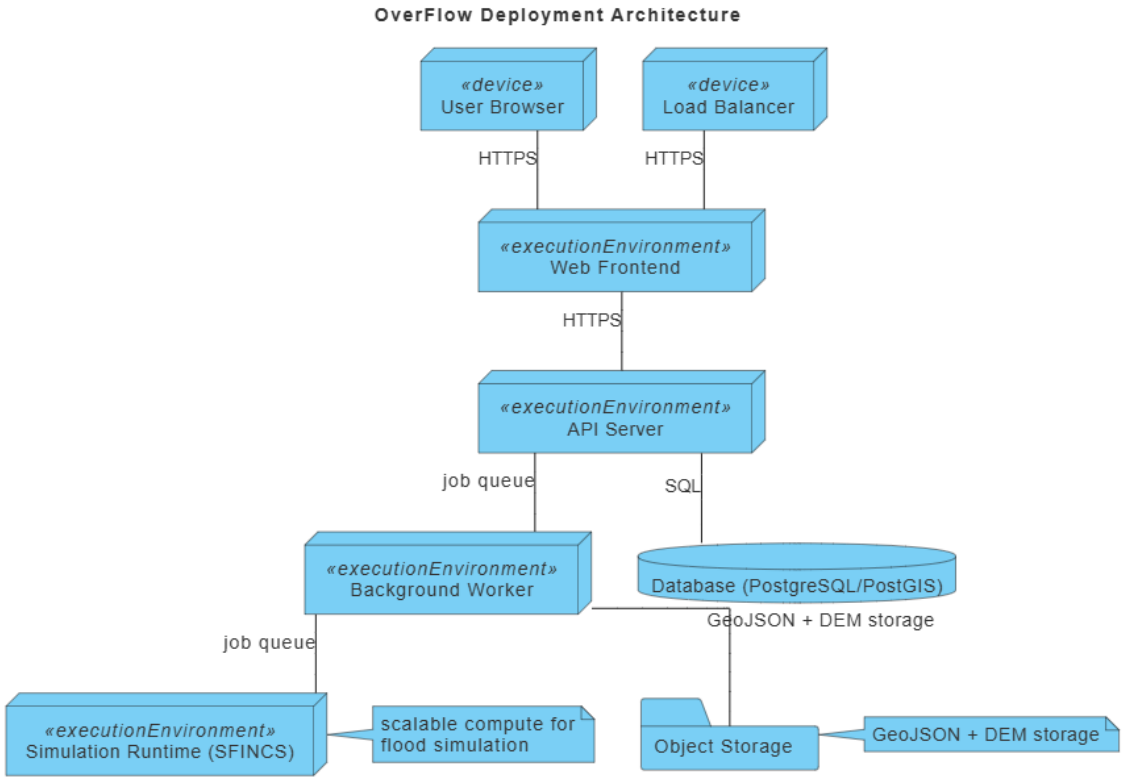


Figure 4.6: Proposed deployment

assignments, and unserved-demand reports. Road exposure shown by a flood layer prompts review; the planner determines which road segments to close before routes are recalculated.

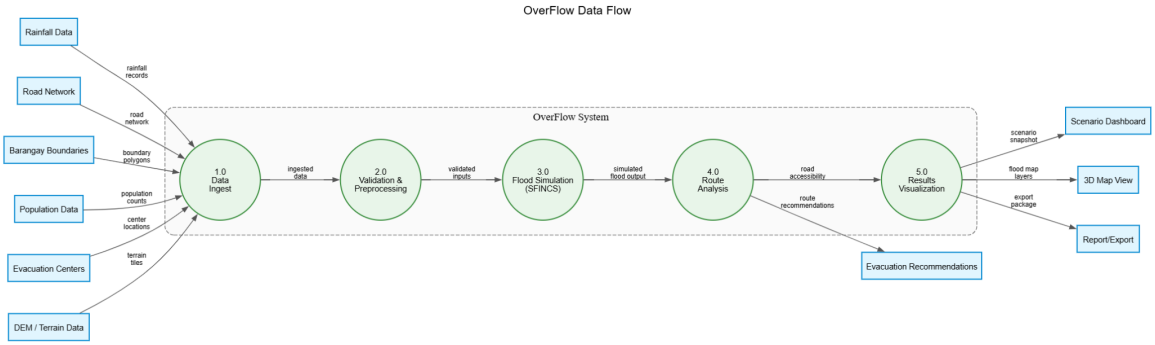


Figure 4.7: Proposed data flow

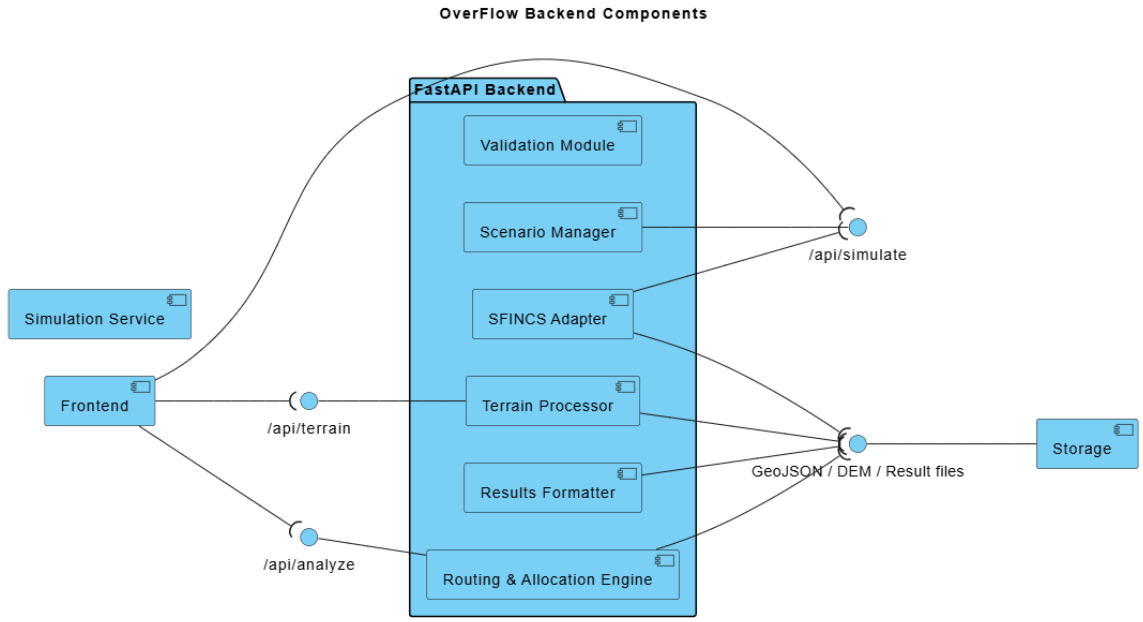


Figure 4.8: Proposed backend components

The activity diagram in Figure 4.9 summarizes the planner's decisions from entering a scenario to reviewing and comparing its outputs. It includes the opportunity to revise inputs when roads are inaccessible or evacuation areas cannot accommodate the entered demand.

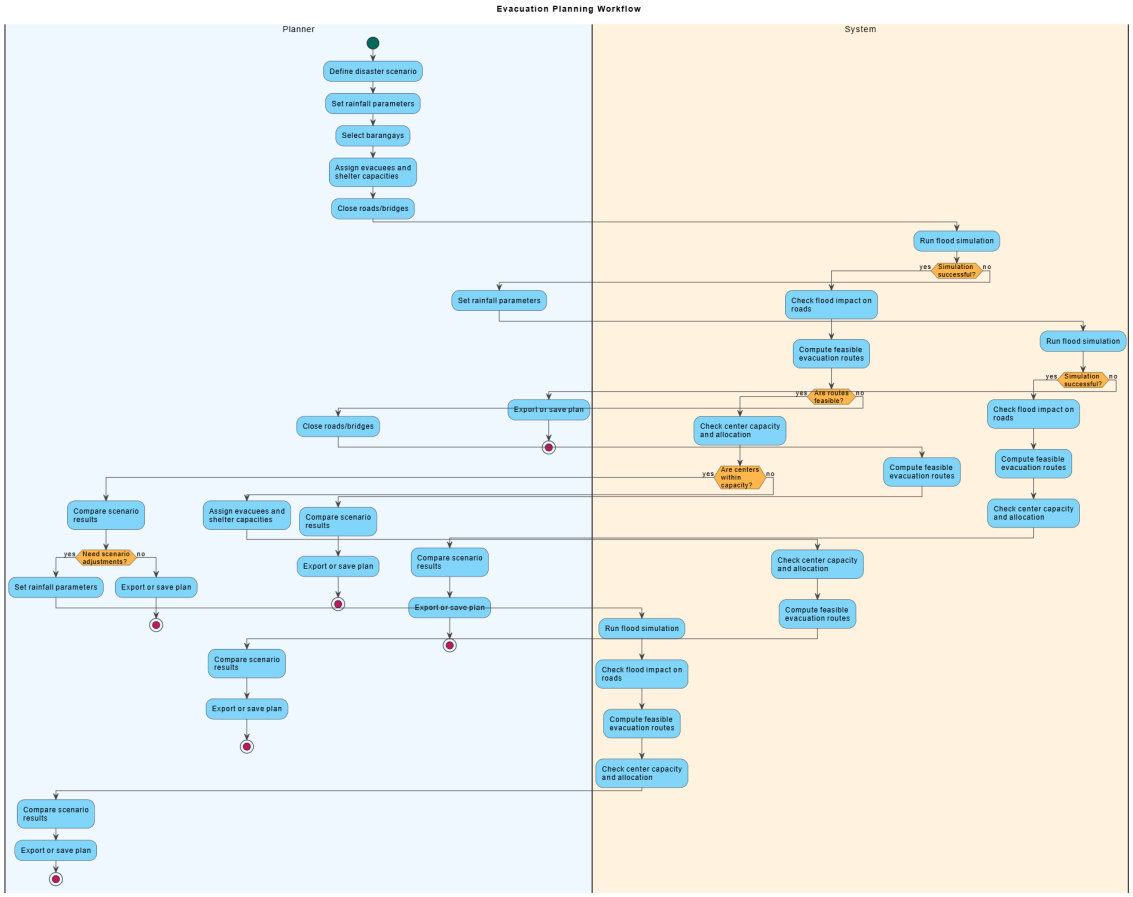


Figure 4.9: Proposed planning activity

Figure 4.10 illustrates the intended three-dimensional map and results dashboard. Its displayed values are placeholders.

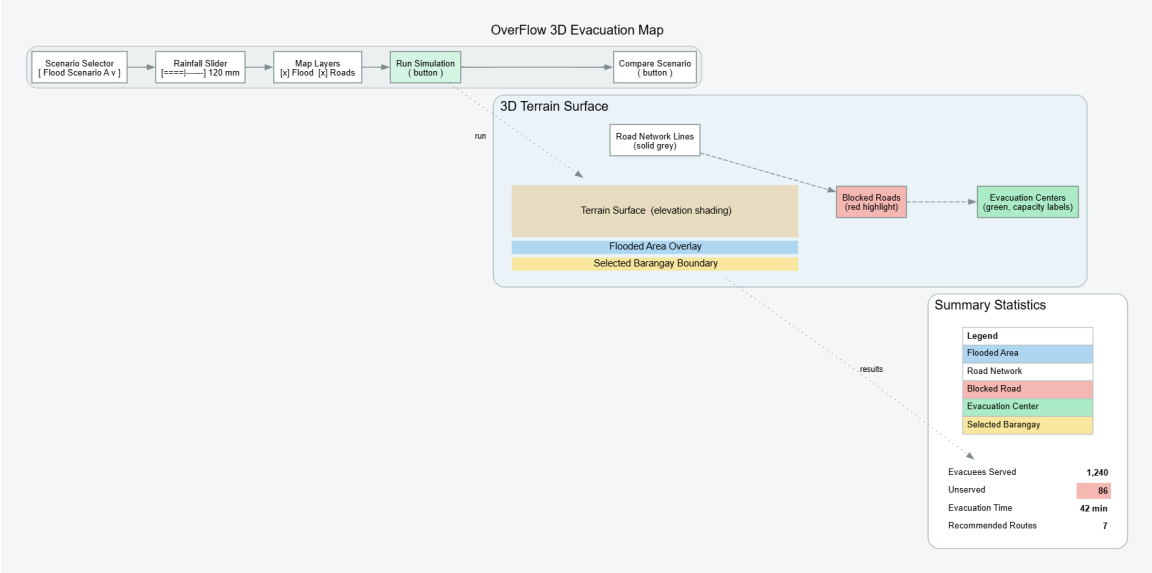


Figure 4.10: Illustrative dashboard mockup; all displayed metrics are placeholders

4.2 Initial Prototype

The prototype demonstrates area selection, scenario controls, a three-dimensional terrain view, rainfall and water-depth playback, manual road closures, center allocation, and comparison of results. Its three demo sectors share a synthetic landscape, and the water animation uses a simplified runoff calculation rather than SFINCS. It therefore demonstrates the intended interaction, not Laoag flood or evacuation outcomes.

Eight automated checks passed on the prototype, including tests that closures change routes and capacity limits can leave demand unassigned.

4.3 Preliminary Flood-Data and Model Checks

The Laoag data include barangay boundaries, roads, a DEM, and flood-susceptibility information. The DEM's approximately 91-metre cells and undocumented elevation accuracy limit local flood interpretation.

A review of the Philippine Space Agency's August 31, 2026 SAR layer found possible flood detections intersecting 49 of Laoag's 80 barangays [3]. This layer provides a dated historical reference, but the detections do not establish road passability.

An initial SFINCS test used the supplied DEM to produce a mapped output. Adjusting an assumed boundary resolved a numerical problem in an earlier run. Since the test did not reproduce a documented flood event, it demonstrates the modeling pipeline but does not establish local flood accuracy.

4.4 System Evaluation

Evaluation to date covers the synthetic prototype and preliminary data-processing steps. Functional tests using Laoag data, comparison of SFINCS results with suitable observations, and staff planning tasks remain to be completed.

5 Conclusions and Recommendations

5.1 Conclusion

5.2 Recommendation

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APPENDICES

A Detailed Tables

B Questionnaires